

Performing subnetting for a given IP address range 192.168.10.0/24

- **Standard /24 (No subnetting) - 1 network with 254 hosts.**
- **Splitting into 4 Subnets (Common for dividing a small office).**
- **Splitting into 8 Subnets (Common for smaller departments).**

SCENARIO # 1: NO SUBNETTING

First 3 octets are for the network and last octet is used for hosts that's how the no subnetting or default subnetting works.

Network Address: 192.168.10.0

Subnet Mask: 255.255.255.0

Total Subnets: 1

Total Usable Hosts Per Subnet: 254 ($2^8 - 2 = 254$)

Usable IP Range = 192.168.10.1 to 192.168.10.254

Broadcast Address = 192.168.10.255

SCENARIO # 2: SPLITTING /24 INTO 4 SUBNETS

We have four departments (HR, Sales, IT Admin). So, we need four separate networks. To split networks we need 4 subnets so, we borrow 2 bits from the host portion

Why 2 bits? Because $2^2 = 4$ (We need to split network into 4 subnets)

New Subnet Mask: /26 (255.255.255.192)

Total Hosts Per Subnet: $2^6 - 2 = 62$

Subnets	Network Address	Usable IP Range	Broadcast Address
1	192.168.10.0	192.168.10.1 - 192.168.10.62	192.168.10.63
2	192.168.10.64	192.168.10.65 - 192.168.10.126	192.168.10.127
3	192.168.10.128	192.168.10.129 - 192.168.10.190	192.168.10.191
4	192.168.10.192	192.168.10.193 - 192.168.10.254	192.168.10.255

SCENARIO # 3: SPLITTING /24 INTO 8 SUBNETS

To split networks into 8 smaller networks we borrow 3 bits ($2^3 = 8$) from the host portion

New Subnet Mask: /27 (255.255.255.223)

Total Hosts Per Subnet: $2^5 - 2 = 30$

Subnet	Network Address	Usable IP Range	Broadcast Address
1	192.128.10.0	192.168.10.1 - 192.168.10.30	192.168.10.31
2	192.168.10.32	192.168.10.33 - 192.168.10.62	192.168.10.63
3	192.168.10.64	192.168.10.65 - 192.168.10.94	192.168.10.95
4	192.168.10.96	192.168.10.97 - 192.168.10.126	192.168.10.127
5	192.168.10.128	192.168.10.129 - 192.168.10.158	192.168.10.159
6	192.168.10.160	192.168.10.161 - 192.168.10.190	192.168.10.191
7	192.168.10.192	192.168.10.193 - 192.168.10.222	192.168.10.223
8	192.168.10.224	192.168.10.225 - 192.168.10.254	192.168.10.255